

Privacy Concerns on Mobile Applications for Google Play Store Market

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Abstract

Since the introduction of smartphones and mobile applications, privacy concerns have been rapidly rising. Today, over one billion people use smartphones. Millions of different apps are being used and installed on daily basis. Whether it's Android, IOS, Windows, or any other mobile operating system, an app might need to request extra permissions to access more data on the user's phone. In this study, Android apps from Google Play Store are examined. Python scraping code was used to collect details about over 5,000 apps including different parameters such as permissions requested, number of installs, reviews, and several other details. The data was then analyzed using SPSS. The analyses made included descriptives and correlation. The data analysis was fruitful and different hypothesis were supported at high significance. The results showed that number of permissions requested by an app is correlated with several other variables such as number of reviews, number of installs, and review score. Limitations are mentioned in the conclusion.

1. Introduction

The number of applications available on Google Play Store has increased very fast in the last years and several studies presented in the literature for this issue [1-17]. Of course, this dramatic increase comes in parallel with the rapidly growing adaptation of smartphones. In 2015, Ericsson, a Swedish multinational telecommunication company, stated that the last decade has also witnessed a stable growth for smartphone users, who by 2020 would be 70% of the world's population. However, Felt et al. (2012) found that no more than 17% of the users actually give attention to what permissions are requested by applications when installed, and up to 42% are even unaware of the concept itself and base their decisions on the reviews and installs count [10]. With such rapid development, concerns are raised regarding users' privacy and the extent of access being granted to apps with permissions.

There are more than 200 different permissions an Android application can request from user. In its documentation, Android classifies the permissions into 4 Protection levels

- Normal (ex: BLUETOOTH – ACCESS NETWORK STATE)
- Signature (ex: MANAGE DOCUMENTS – READ VOICEMAIL)
- Dangerous (ex: READ PHONE NUMBERS – SEND SMS)
- Special (ex: SYSTEM ALERT WINDOW – WRITE SETTINGS)

According to the permission type, the user will be notified when first opening the app for the first time. A pop message will

show asking to grant access to certain data according to permissions type like what is shown in image 1 below.

Scholars and research centers tried to examine different aspects of the “Apps Permissions” issue such as awareness and behavior (Felt et al., 2012) [10], types of permission and risk implications [3,5], or suggestions of technical improvements [4,7, 9].

The main objective of this study is to examine the relation between some variables of an app on the Google Play Store and the number of permissions requested when installing it. Since some previous studies pointed at the behavior of referring to reviews and installs count instead of permissions when installing, the focus was set on the relatively related variables.

The study is mainly composed of four classic sections. The first one is this specific introduction for the study. Then, the methodology is explained in detail for all phases: coding the Python script, preparing the categories links, scrapping / collecting the data, cleaning and proofing the data, and statistical analysis using different software. Later the findings of all tests and measurements such as correlation, regression, and ANOVA are presented. Finally, findings are interpreted, and research limitations are briefly specified and explained.

In this study, results were deeply discussed and supervised. Results showed that number of permissions had a significant relationship with all other variables and impact on them based on multiple regression testing.

In this version, the simple model is displayed with all hypotheses tested without discussing all details of modeling:

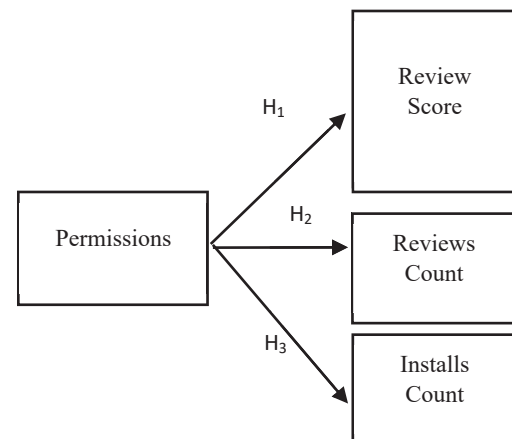


Fig. 1: Theoretical framework

2. Model Variables

Permissions: In the model, 'Permissions' variable is the cardinality (count / number) of permissions requested by the app from the user. In the app store, permissions are shown as string and classified under different types such as: Storage, Wi-Fi connection information, and others. However, in the methodology section, there will be an explanation on how the cardinality was obtained. Permissions, and their cardinality, requested are the main variable of this study. The analysis aims at understanding the independent effect and relationship of this main variable with other variables.

Review Score: In the model, 'Review Score' variable is the average score of reviews made for a specific app using a '1-to-5' scale with 0.1 increments. The score is calculated based on the Play store users.

Reviews: In the model, 'Reviews' variable is the cardinality (count / number) of reviews made on a specific app. The number on the store page is exact with no rounding like in the case of other variables such as number of installs which will be shown in the coming sub-section.

Installs Count: In the model, 'Installs Count' variable is the cardinality (count / number) of installs made from a specific app. However, it is important to note that this number is ordinal/categorical. The indicator shows the record broken by the app instead of actual number of installs (such as: 5000+ instead of 5,435 or 1.7M instead of 1,700,956). This limitation was minimized in data cleaning.

Hypotheses:

- H 1: 'Permissions' (number of permissions) has a significant impact on 'Review Score'.
- H 2: 'Permissions' (number of permissions) has a significant impact on 'Reviews' (number of reviews).
- H 3: 'Permissions' (number of permissions) has a significant impact on 'Installs Count'.
- H 4: there is a significant relationship between 'Permissions' (number of permissions) and 'Review Score'.
- H 5: there is a significant relationship between 'Permissions' (number of permissions) and 'Reviews' (number of reviews).
- H 6: there is a significant relationship between 'Permissions' (number of permissions) and 'Installs Count'.
- H 7: there is a significant relationship between 'Permissions' (number of permissions) and 'Year Updated'.
- H 8: there is a significant relationship between 'Permissions' (number of permissions) and 'Reviews/Installs' (ratio of reviews number / number of installs).
- H 9: there is a significant relationship between 'Reviews/Installs' (ratio of reviews number / number of installs) and 'Year Updated'.
- H 10: there is a statistically significant difference among 'Developer' with respect to 'Permissions'.
- H 11: there is a statistically significant difference among 'Category' with respect to 'Permissions'.
- H 12: there is a statistically significant difference among 'Year Updated' with respect to 'Permissions'.

3. Methodology

In this section, a detailed technical methodology will be explained regarding coding the scraping Python script, choosing the sample (categories / links) of the apps, data scraping process, data proofing and cleaning, and statistical analysis.

Coding – Python

This section is technical and requires fundamental knowledge regarding coding and programming. The main language used was Python using version 2.7.10 operated on MAC OS MOJAVE 10.14.5. In the Python code, Selenium library was imported and used to be able to control internet browser. The code requires and works along with JS – JavaScript, Node, and ChromeDriver (since Google Chrome was chosen).

The code would first get 3 parameters from the user on the terminal / command line: location / directory of the chrome driver, URL of the apps category needed to be scrapped, and, optionally, an integer indicating the number of scrolls (will be explained); if the number of scrolls is not indicated, the code scrolls down enough to load all possible apps. Then, when the code is executed, the Google Chrome web browser will launch and go to the indicated URL. The browser was first cleared from cookies and any user's data to avoid suggestions and recommendations. Only IP address was an affecting factor which relatively displayed more Turkish local apps (the location of the study was Istanbul). The first set of apps will be loaded, and the code will start scrolling down to stimulate the browser to load more apps at the bottom of the page (based on the third parameter). When the apps are loaded, the code scraps/collects all the apps' Id's and initiate a CSV file using CSV library imported. The CSV file will start with columns-based parameters needed to be collected. First column would be app id. After it comes any parameters needed. To specify the parameters, 'Container Id' was used. Container Id were previously recorded for the needed parameters. The collected parameters in this study were:

- ID: App unique Id (ex: com.whatsapp)
- Name: App Name (ex: WhatsApp)
- Category: the category which the app belongs to (ex: Communication)
- Reviews: the number of reviews made on the app (ex: 135,846 reviews)
- Installs: the number of installs made from the app (ex: 35M)
- Last Updated: the date of last updated released (ex: 25/01/2019)
- Developer Name: the name of developer or developing company (ex: Whatsapp Inc.)
- Permissions: all the permissions requested by the app from the user delimited with comma ',' (ex: access to contacts, access to Wi-Fi status)

The last parameter, 'Permissions', required an extra step to be collected as it was the only parameter not found in the page and required opening a link "detail". In the code, Try-Catch / Try – Except blocks were needed. If the needed parameter or specified container was not found, error was printed on the screen. In the case of missing detail, the code was designed to simply skip the record and move to the next one. However, in the case of wrong container, usually due to opening from different IP address (different IP addresses may reach different page designs), the container name was checked, updated, and the code was rerun. The code also was designed to display the number of apps found and number of apps successfully collected.

When the process was over, the CSV file would be found in the same directory of the script.

4. Preparation of Categories – Choosing Sample

In Google Play Store, there are several classifications and categorizations for the apps. In the homepage, even if the user is visiting for the first time, the store recommends some apps. Also,

top rated games, popular games, and other top charts are displayed.

Since number of installs, number of reviews, and review score were involved in the study, all top charts and recommendations were avoided. Only categorical charts were used which rely only on the nature of the app (ex: Games > Action, Business). Scrapping was determined by displayed order ('Daydream' chart till 'Word' chart under 'Games'). 'Family' chart was not scraped as the size needed for the sample was achieved (5000+ record). However, in some charts, there were sub-charts with top charts included. All top charts were ignored.

A total of 39 categories / subcategories for 4,805 apps (5,264 app before data cleaning / proofing) were scrapped.

The scraping was done between 3 June, 2019 and 20 July, 2019. A total of 52 CSV files were obtained by six iterations (5 iterations x 10 parallel tabs + 2 tabbed iteration). The CSV files were then merged using a terminal command. Final CSV file was ready to be cleaned and processed.

The following analyses were made:

Descriptive Analysis: A general Descriptive analysis was made using SPSS for the collected parameters stating their 'N', Mean, Median, Mode, Std. Deviation, Variance, Minimum, and Maximum. Categorical variables (Category and Developer) were processed separately due to statistical and string-processing limitations.

Correlation Analysis: Correlation analysis was made using SPSS to examine the existence and strength of relationship between variables mentioned in the hypotheses: Permissions and Review Score - Permissions and Reviews Count - Permissions and Installs Count. Moreover, 'Year updated' and the suggested ratio of 'Reviews/Installs' were added to the correlation and all correlations were examined.

However, as the results will show, both normality and homoscedasticity assumptions were violated. Hence, 'Robust Tests of Equality of Means' analysis was made using the conservative approaches of 'Welch' and 'Brown-Forsthe' where the results were valid and 'Games-Howell' approach was then applied for the multiple comparison.

5. Findings

Descriptive Findings: Descriptive findings from the SPSS are shown below in Table 1. The variables are respectively:

		Reviews	ReviewScore	InstallCount	RevInsRatio	Yearupdated	perms
N	Valid	4805	4805	4805	4805	4805	4805
	Missing	0	0	0	0	0	0
Mean		407942.72	4.288	24528306.97	.024007084966868	2018.76	12.42
Median		25537.00	4.300	1000000.00	.013237000000000	2019.00	10.00
Mode		121 ^a	4.5	1000000	.006260000000000	2019	7
Std. Deviation		2897528.629	.3320	193390448.773	.033461556712170	.759	8.094
Variance		8395672157834.840	.110	37399865676707400.000	.001	.575	65.517
Minimum		100	2.2	5000	.000009920000000	2011	0
Maximum		92888725	5.0	5000000000	.554374000000000	2019	133
a. Multiple modes exist. The smallest value is shown							

Table 1: Descriptive Analysis Results for Scale Variables

'Reviews' for the number of reviews on an app, 'ReviewScore' for the review score on an app, 'InstallCount' for the number of installs made from an app, 'RevInsRatio' for the ratio of reviews to installs, 'Yearupdated' for the year of last update, and 'perms' for the number of permissions requested by an app. The total number of apps after cleaning was 4,805 with no missing data.

Looking at table 1 we can deduce that there is a high number of reviews and install count for the applications within this dataset. Also, looking at the mean of the ReviewScore column, it can be understood that applications are mainly 4-star applications out of the maximum 5. The Yearupdated column gives us the last year the application was updated. Looking at the mean, most applications have been updated within the year 2018. Lastly, the perms table gives the number of permissions for the applications 0 being the minimum and 133 the maximum within the dataset. Again, looking at the mean value of this column it can be said that the applications are asking for an average of 12 permissions from users.

Figure 2 shows a graph created on PowerBI, a visualization tool. This graph shows, decreasingly, the number of permissions requested by applications from different categories. The permissions are color coded as low, medium, and critical privacy. An interesting find can be seen when looking at the photography category. This category of applications not only requests the biggest number of permissions, but also requests the highest number of critical permissions.

For correlation analysis, hypotheses H4 to H12 were tested to check the significance of relationship between different variables. Table 2 below shows all the correlation results from SPSS output.

As shown in the table, all relationships are significant EXCEPT:

- Review Score and Installs Count
- Installs Count and Reviews to Installs ratio

While examining the correlations results, it is noticed that 'Permissions' are significantly correlated with all other variables with a value up to 0.191 (Permissions and Installs Count).

Hypotheses results:

All Hypothesis were significantly supported.

● Low Privacy ● Medium Privacy ● Critical Privacy

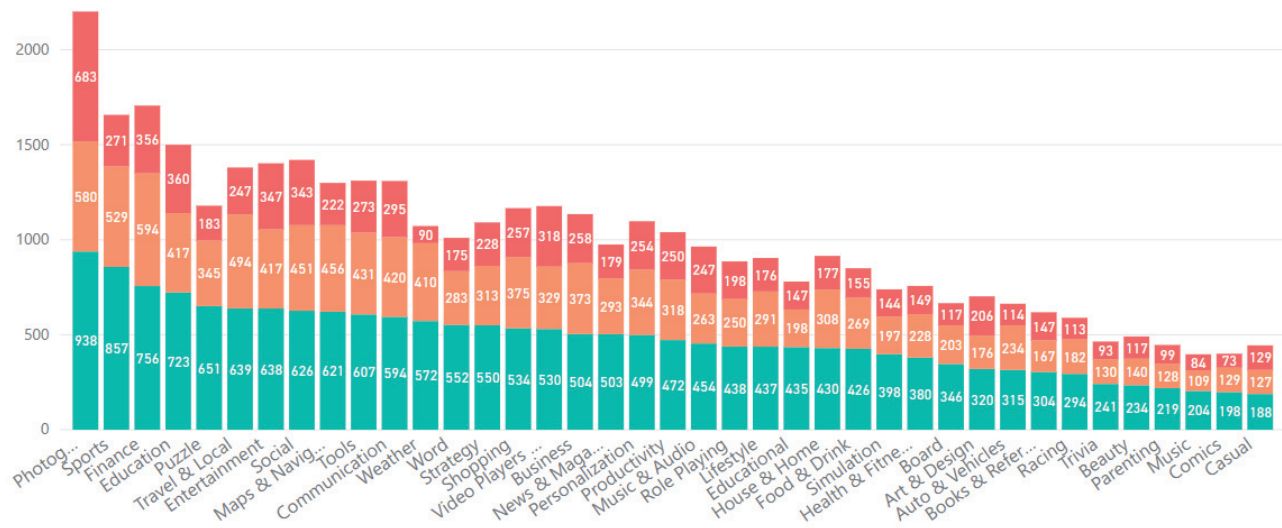


Fig. 2: Number of permissions requested by application categories

Correlations							
		Reviews	ReviewScore	InstallCount	RevInsRatio	Yearupdated	perms
Reviews	Pearson Correlation	1	.048**	.389**	.139**	.033*	.176**
	Sig. (2-tailed)		.001	.000	.000	.023	.000
	N	4805	4805	4805	4805	4805	4805
ReviewScore	Pearson Correlation	.048**	1	.010	.317**	.140**	-.095**
	Sig. (2-tailed)	.001		.506	.000	.000	.000
	N	4805	4805	4805	4805	4805	4805
InstallCount	Pearson Correlation	.389**	.010	1	-.028	.023	.191**
	Sig. (2-tailed)	.000	.506		.053	.118	.000
	N	4805	4805	4805	4805	4805	4805
RevInsRatio	Pearson Correlation	.139**	.317**	-.028	1	.082**	.046**
	Sig. (2-tailed)	.000	.000	.053		.000	.001
	N	4805	4805	4805	4805	4805	4805
Yearupdated	Pearson Correlation	.033*	.140**	.023	.082**	1	.154**
	Sig. (2-tailed)	.023	.000	.118	.000		.000
	N	4805	4805	4805	4805	4805	4805
perms	Pearson Correlation	.176**	-.095**	.191**	.046**	.154**	1
	Sig. (2-tailed)	.000	.000	.000	.001	.000	
	N	4805	4805	4805	4805	4805	4805
**. Correlation is significant at the 0.01 level (2-tailed).				*. Correlation is significant at the 0.05 level (2-tailed).			

Table 2: Correlation values from SPSS

6. Conclusions

As seen in the results of the analyses done, the model stands valid against all the tests. The variable 'Permissions' that stands for the number of permissions requested by an app from user, is correlated with the other variables:

- Reviews: number of reviews
- Review Score: review score of the app based on users' reviews
- Year Updated: the year of the last updated made
- Installs Count: the number of installs made from the app

- Reviews/Installs Ratio: the ratio of number of reviews made to number of installs, stating the rate of reviewing an app by installers.

- For this study, there were several limitations that had affected the results directly and indirectly. Although 5,264 apps is statistically big enough to conduct such analyses, the sample size is small compared to the huge number of apps on Google Play Store, over 3 million apps.

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